

Integrated Modeling and Control Optimization of Biped Wheel-Legged Robot

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Abstract—Biped wheel-legged robot is an important configuration of ground mobile robot. In the existing research, in order to deploy the model on an embedded system with limited computing power, the balance control of the robot is usually decoupled from the body posture and steering, which reduces the control coordination of the robot. In order to solve the above problems, firstly, the novel Full-State Dynamics Model(FSDM) is introduced, and the model is linearized by Taylor expansion and solving the limit of multivariate function. Secondly, a novel Forward Kinematics(FK) solution method based on trajectory equation is proposed for Virtual Model Control(VMC). Compared with the general FK solution method, it can further significantly improve the calculation speed of VMC on the embedded platform. Furthermore, the Linear Quadratic Regulator(LQR) controller is optimized, and the weight matrix value can be automatically adjusted according to the error of the state variable. At the same time, simulation results show that the motion performance of the robot can be improved by actively adjusting the posture. Therefore, an adaptive LQR controller, a steering compensator and a gravity compensator are designed. Simulation and physical experimental results verify the effectiveness of the proposed model, controller and control strategy.

Index Terms—Wheeled robots, motion control, dynamics, kinematics, optimization and optimal control.

I. INTRODUCTION

WHEEL-LEGGED robots are a class of mobile robots that combine the high speed and efficiency of wheeled robots with the robust obstacle-crossing capabilities of legged robots [1], [2], having great potential in site inspections, military tasks, and other scenarios. Among wheel-legged robots, bipedal wheel-legged robots are ideal platforms for integrating components such as LiDAR, robotic arms, and upper-body humanoid

robots due to their compact structure. According to the leg structures, bipedal wheel-legged robots can be classified into serial legs and parallel legs. Serial-legged robots have a relatively simple structure with high degrees of freedom and flexibility. Typical examples include the Handle [3] developed by Boston Dynamics and WLR series developed by Harbin Institute of Technology [4], [5]. Compared to parallel-legged structures, serial-legged robots typically exhibit lower load capacity. Parallel kinematics chains typically exhibit an n-link architecture, with four-link and five-link structures being the most prevalent in current research and applications. The four-link configuration, while eliminating the knee joint motor and offering good flexibility and adaptability, also presents several challenges, such as uneven load distribution and higher material strength requirements. These issues are evident in the Ascento robot developed by ETH Zurich [6], [7]. In contrast, the five-link structure offers better balance and load capacity, making it suitable for tasks requiring high stability, exemplified by Tencent's Ollie robot [8], [9].

The leg configuration in bipedal wheel-legged robots directly impacts the complexity of modeling and control. The newton-euler or lagrangian dynamics modeling methods are widely applied in such studies [10], [11], [12], [13], [14], [15]. To simplify the modeling process, researchers have proposed using Virtual Model Control(VMC) to independently model the legs. In the dynamics model, the structure of each real leg is treated as an equivalent virtual leg [13]. By using FK equations and the principle of virtual work, the mathematical relationship between joint torques and end-effector forces is derived, thereby obtaining the dynamics model of the legs. Nevertheless, the VMC method requires the calculation of the FK Jacobian matrix of the leg. If the FK equations are solved using geometric methods, they will involve numerous trigonometric functions and exponentiation. This results in highly complex expressions, requiring the MCU to handle a large number of computations, which reduces the real-time performance of the control system and makes it difficult to deploy the model in embedded systems with limited computational resources. In terms of control strategies, some research have only designed model-based controllers for the balancing phase of the robot, while steering motion and support phase are decoupled [15], [16], [17]. However, when the robot is required to execute complex commands involving multiple states, significant incoordinated vibrations will occur, which greatly reducing the dynamics response performance of the robot. To address the aforementioned issues, the Full-State Dynamics Model (FSDM) that encompasses all motion states of the robot is proposed. The design of steering and gravity

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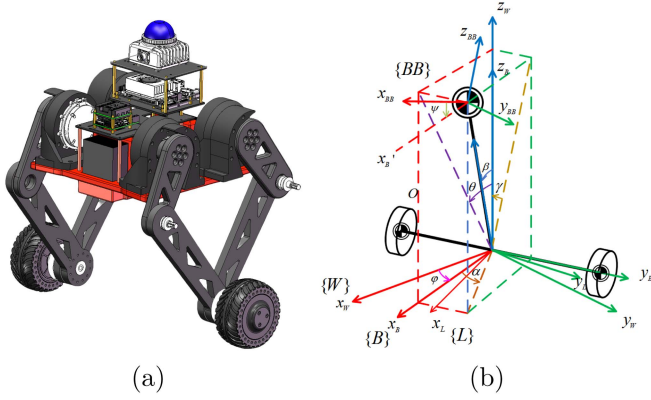


Fig. 1. Biped wheel-legged robot (a) Physical model. (b) Description of the coordinate systems.

controllers is eliminated. The FSDM and compensators effectively avoid the control incoordination caused by decoupled control strategies. To increase control frequency and reduce Microcontroller Unit(MCU) computational load, a novel VMC solving method based on trajectory equation is introduced. Additionally, to further improve the dynamics response performance of robot, an Adaptive LQR algorithm based on parameter adaptation is designed. The main contributions of this article can be summarized in the following aspects:

- 1) A dynamics model including all motion states of the robot is established to facilitate determination of control parameters.
- 2) A method for calculating the FK equation based on the circular trajectory equation is proposed, which simplifies the computational complexity.
- 3) An Adaptive LQR control algorithm and an optimized control strategy are designed to further improve the motion performance of the robot.

In this paper, the proposed model and theoretical analysis are verified by simulation and deployed on the prototype. The simulation results show that under the same parameter conditions, compared with the simplified model, FSDM and control strategy improve the response speed of the biped wheel-legged robot.

II. DYNAMICS MODELING

The symmetrical five-link structure is deployed in the robot, and the modeling complexity is simplified by integrating VMC. To establish the FSDM, the commonly used planar wheel-inverted pendulum model is extended to three-dimensions, as shown in Fig. 1.

To reduce unnecessary model complexity and simplify linearization, the following assumptions are made in this research:

- 1) No frictional damping between the body and the leg components.
- 2) A gravity feedforward compensator has been established, and the gravity matrix method is omitted in the lagrangian dynamics equations.
- 3) The Coriolis force generated by the leg is ignored.

TABLE I
PARAMETERS DEFINITION

Symbol	Definitions
m_b, m_w	Body and pendulum mass, wheel mass
r, D, l_b	Wheel radius, Body width, The predefined pendulum length in the dynamics model
$x, \dot{x}, l, \dot{l}$	Distance, Forward velocity, Pendulum length, Pendulum length rate
$\psi, \dot{\psi}, \theta, \dot{\theta}$	Body's pitch and its rate, leg's pitch and its rate
$\varphi, \dot{\varphi}, \gamma, \dot{\gamma}$	rate, yaw and its rate, roll and its rate
α, β	The angle between the projection of the pendulum in $x_B O y_B$ and x_B , the angle between the pendulum and z_B
$I_{bx}, I_{by}, I_{bz}, I_{wx}, I_{wy}, I_{wz}$	Body's moment of inertia in $\{BB\}$, wheel's moment of inertia in $\{B\}$
T_L, T_R	Left and right drive wheel torque
T_{pL}, T_{pR}	Left and right virtual leg torque
F_L, F_R	Left and right virtual leg normal force
$\Delta l_L^*, \Delta l_R^*$	The change value of left and right leg length when the robot actively adjusts the roll angle
f_L, f_R	The axial friction force of the left and right wheels when the robot is turning
F_{gL}, F_{gR}	The output value of the gravity compensator of the left and right legs

A. Full-State Dynamics Model

A dynamics model based on the full-state of the robot is established by using the lagrangian dynamics modeling method. The model realizes the unified description of the posture and motion state. By actively controlling the body posture, the dynamics performance of the robot is improved, thereby enhancing its stability and adaptability in complex environments. For convenience, multiple coordinate frames are established on the robot, as shown in Fig. 1(b). The reference is set based on the robot's stationary equilibrium posture in the world coordinate system $\{W\} = [x_W, y_W, z_W]$. The body coordinate system $\{B\} = [x_B, y_B, z_B]$ is obtained by rotating around the z_W -axis of the world coordinate system by an angle φ , which represents the body coordinates after steering. The z_L -axis of the pendulum coordinate system $\{L\} = [x_L, y_L, z_L]$ coincides with the pendulum and moves with its motion. The body reference coordinate system $\{BB\} = [x_{BB}, y_{BB}, z_{BB}]$ aligns with the robot's azimuth. These coordinate systems are interrelated through transformation matrices, and the velocity vectors of different parts of the robot are expressed in these frames. For detailed expressions, please refer to Section I in the supplementary materials. The parameters defining the primary models of the system are listed in Table I.

The pendulum length l_b in the model corresponds to the physical meaning of the fuselage height in practice. It should be a variable but for simplicity, it is predefined as a constant. In practical applications, according to pendulum length l , several feedback matrices are calculated for the controller to call.

In addition, in the actual control, the left and right virtual legs are controlled by independent VMC controllers, so the pendulum length l will be determined according to the length of the virtual legs on both sides. The lagrangian dynamics equations are as follows:

$$T_{trans} = \frac{1}{2}m_w(v'_L \cdot v_L + v'_R \cdot v_R) + \frac{1}{2}m_b(v'_b \cdot v_b) \quad (1)$$

$$T_{rot} = \frac{1}{2}m_w(w'_L I_w w_L + w'_R I_w w_R) + \frac{1}{2}m_b(w'_b I_b w_b) \quad (2)$$

$$L = T_{trans} + T_{rot} - V \quad (3)$$

$$V = \frac{1}{2}m_b g l_b \cos(\beta) \quad (4)$$

Where T_{trans} represents the translational kinetic energy of the robot, T_{rot} denotes the rotational kinetic energy of the robot, and V is the potential energy of the robot. Define the generalized coordinates $q = [x, \psi, \theta, \varphi, \gamma, l]^T$, and establish the lagrangian equations:

$$\frac{d}{dt} \left(\frac{\partial \dot{L}}{\partial \dot{q}} \right) - \frac{dL}{dq} = Q_i \quad (5)$$

The resulting equations constitute a relatively accurate dynamics model, with complex expressions that include numerous nonlinear terms and instances of multivariable coupling. Nevertheless, in practical control scenarios, these nonlinear terms often yield only minimal values. Therefore, linear controllers can still be considered.

B. Model Linearization

Due to the large number of coupling terms in the model, the system description is highly complex. To use a linear controller, further mathematical processing is required. Yet, a first-order Taylor expansion cannot linearize the coupling terms completely in this model. Considering that in practical scenarios, ψ , θ and γ are all minimal quantities, and that ψ has a higher priority in control. Therefore, the approach is as follows:

After the first-order Taylor expansion of the model, limits are first taken at $\psi = 0$, then θ and γ are treated as infinitesimal quantities of the same order, and limits are taken for the multivariable functions. After that, the linearized dynamics model is obtained as follows:

$$\begin{cases} a_1 \ddot{x} + a_2 \ddot{\theta} = c_1 T_L + c_2 T_R \\ a_3 \ddot{\psi} + a_4 \ddot{\theta} = c_3 T_{pL} + c_4 T_{pR} \\ a_5 \ddot{x} + a_6 \ddot{\psi} + a_7 \ddot{\theta} + a_8 \theta = c_5 T_L + c_6 T_R \\ a_9 \ddot{\varphi} = c_7 T_L + c_8 T_R \\ a_{10} \ddot{\gamma} + a_{11} \gamma = c_9 F_L + c_{10} F_R \\ a_{12} \ddot{l} = c_{11} F_L + c_{12} F_R \end{cases} \quad (6)$$

Where $a_1 = m_b + 2m_w + \frac{2I_{wy}}{r^2}$, $a_2 = a_5 = m_b l_b$, $a_3 = a_4 = I_{by}$, $a_7 = I_{by} + m_b l_b^2$, $a_8 = a_{11} = -\frac{m_b g l_b}{2}$, $a_9 = I_{bz} + 2I_{wz} + \frac{m_w d^2}{2} + \frac{I_{wy} d^2}{2r^2}$, $a_{10} = (I_{bx} + 2I_{wx} + 2m_w r^2 + m_b l_b^2)$, $a_{12} = m_b$, $c_1 = c_2 = \frac{1}{r}$, $c_3 = c_4 = 1$, $c_5 = c_6 = -1$, $c_7 = -c_8 = -\frac{d}{2r}$, $c_9 = -c_{10} = \frac{d}{2}$, $c_{11} = c_{12} = 1$

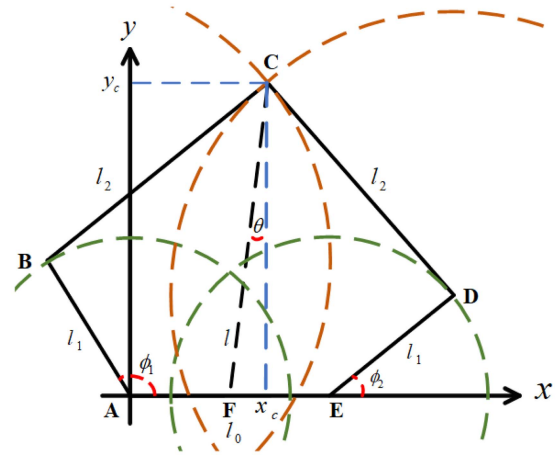


Fig. 2. Motion trajectory diagram of five-link structure.

From the perspective of the linearized state-space equations, the control of some variables can be realized by different controllers. However, in practical applications, using different controllers may lead to insufficient coordination of motion control, thereby reducing the dynamics response performance of the system. Therefore, these variables still need to be integrated into the same controller. Select the state variables $\varepsilon = [x, \dot{x}, \psi, \dot{\psi}, \theta, \dot{\theta}, \varphi, \dot{\varphi}, \gamma, \dot{\gamma}, l, \dot{l}]^T$, and the inputs $u = [T_L, T_R, T_{pL}, T_{pR}, F_L, F_R]^T$, and establish the state-space equations:

$$\dot{\varepsilon} = A\varepsilon + Bu \quad (7)$$

Where A is a 12×12 matrix, B is a 12×6 matrix. Through the above method, the linearized dynamics model is established.

C. FK Based on Trajectory Equation in VMC

In order to simplify the dynamics model of parallel-legged structure, the virtual legs is defined as the intermediate medium between the dynamics model of the wheeled inverted pendulum and the real structure, Through the VMC method, the mapping between the force/torque of the virtual leg and the joints torques is obtained by solving the FK equation and the virtual work principle. The virtual leg is used, which simplifies the complexity of the robot dynamics model. As mentioned in the literature [13], the general FK solution is usually realized by the geometric relationship between the linkages.

Since the general FK equation involves a large number of nested operations of trigonometric functions and power exponents, the process of partial differential and the solution of the equation are very complicated. To address this, A method for calculating the FK equation based on the circular trajectory equation is proposed. In the plane of the leg links mechanism, as shown in Fig. 2, the coordinate system is established with the hip joint A as the origin of the coordinate system and the link between joint A and joint B as the x-axis.

The distance between joint A and joint E is defined as $AE = l_0$, and F is the midpoint of AE. The thigh length is defined as $AB = ED = l_1$, and the calf length is defined as $BC = DC =$

l_2 , Let ϕ_1 be the angle between AB and the positive x -axis, and ϕ_2 be the angle between ED and the positive x -axis. From this, it can be seen that the trajectory of the thigh and calf is circular, which can be expressed by the circular trajectory equation in analytic geometry:

$$\begin{cases} (x - l_1 \cos(\phi_1))^2 + (y - l_1 \sin(\phi_1))^2 = l_2^2 \\ (x - l_0 - l_1 \cos(\phi_2))^2 + (y - l_1 \sin(\phi_2))^2 = l_2^2 \end{cases} \quad (8)$$

It can be observed in Fig. 2 that there are two intersections of the two calf trajectories. By analyzing the expressions for y_c in the two intersections, it is found that y_{c1} must be smaller y_{c2} due to $\phi_1 \in [\frac{\pi}{2}, \frac{7\pi}{6}]$ and $\phi_2 \in [-\frac{\pi}{6}, \frac{\pi}{2}]$. Through geometric configuration analysis, it can be inferred that the intersection containing y_{c1} should be discarded, leaving only one effective intersection. The specific equations can be found in section II of the supplementary materials. The FK equation can be simply expressed as:

$$\begin{cases} x_c = f_1(\phi_1, \phi_2) \\ y_c = f_2(\phi_1, \phi_2) \end{cases} \quad (9)$$

By using the FK equation obtained by the circular trajectory equation, the nested operation of trigonometric functions and power exponent is effectively avoided, and the computational complexity of the Jacobian matrix and polar coordinates of point C is significantly reduced. As for the final expression, many intermediate variables can be reused in code writing, thereby reducing the calculation time in a control period and increasing the maximum control frequency of the controller.

The Jacobian matrix is obtained by total differential on both sides of the equation, and the mapping relationship between the joint torque and the output of the end-effector of the mechanism is derived based on the principle of virtual work. Total differentiation of the equations:

$$\begin{bmatrix} \delta x_c \\ \delta y_c \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial \phi_1} & \frac{\partial f_1}{\partial \phi_2} \\ \frac{\partial f_2}{\partial \phi_1} & \frac{\partial f_2}{\partial \phi_2} \end{bmatrix} \begin{bmatrix} \delta \phi_1 \\ \delta \phi_2 \end{bmatrix} \quad (10)$$

Simply expressed as:

$$\delta x = J \delta q \quad (11)$$

According to the principle of virtual work, it follows:

$$T = J^T F \quad (12)$$

In addition, the value of the virtual leg is calculated as follows:

$$l = \sqrt{\left(x_c - \frac{1}{2}l_0\right)^2 + y_c^2} \quad (13)$$

$$\theta = \arctan\left(\frac{x_c - 1/2l_0}{y_c}\right) \quad (14)$$

Results of comparative experiments show that, under the same input conditions, the calculation times for the VMC expressions derived from the general kinematics model are 0.007354 s, 0.006646 s, and 0.006554 s, with an average time of 0.006851 s. In contrast, the VMC expressions derived from the trajectory equations take 0.001587 s, 0.001227 s, and 0.001204 s, with an average time of 0.001339 s. The speed of solving FK equation by this method is improved by about 80%.

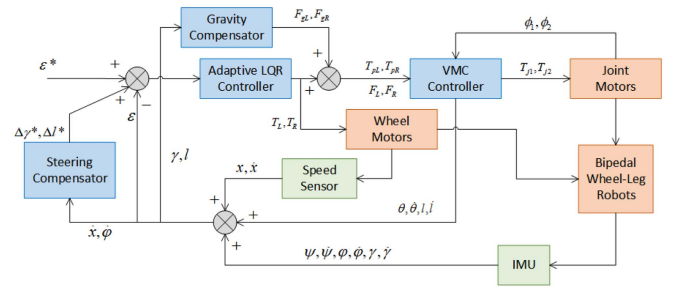


Fig. 3. In order to facilitate the understanding of the control process, the VMC controller part is simplified, and the left and right sides are not distinguished, but it is still distinguished in the actual control.

III. CONTROLLER DESIGN

The LQR is a widely used control approach in optimal control theory, designed to determining the optimal control input by minimizing the quadratic cost function of a linear system. The cost function and control input can be expressed as:

$$\min(J) = \int_0^\infty (\varepsilon^T(t)Q\varepsilon(t) + u^T(t)Ru(t))dt \quad (15)$$

$$u(t) = -K\varepsilon(t) \quad (16)$$

Where K is the control gain, which can be obtained by solving the following Riccati equation:

$$A^T P + P A - P B R^{-1} B^T P + Q = 0 \quad (17)$$

$$K = R^{-1} B^T P \quad (18)$$

The optimal control of the system under this weight can be realized by selecting the appropriate weights matrices and calculating the state feedback gain matrix. However, there is no quantitative method for determining the weights matrices, and the results may not be the optimal in different scenarios. It is expected that weights will be changed based on environment, thereby recalculating the state feedback gain matrix to further enhance the dynamics response performance of the robot. A parameter adaptive strategy for LQR and response optimization compensators are proposed in this article. The control blocks for the robot is shown in Fig. 3.

A. Adaptive LQR Based on Errors

The error-based adjustment strategy does not rely on external information from sensors other than the Inertial Measurement Unit(IMU). It adjusts the weights matrices based on state errors. The weights update expression is:

$$\{Q(t+1) = Q(0) + \text{diag}(\lambda) \cdot \text{diag}(|e(t)|)\} \quad (19)$$

where λ is the adjustment rate, $e(t)$ is the error between feedback and reference. This method can make the control priority respond to the state variables with large error without a lot of operations, and can quickly adjust the weights with good real-time performance.

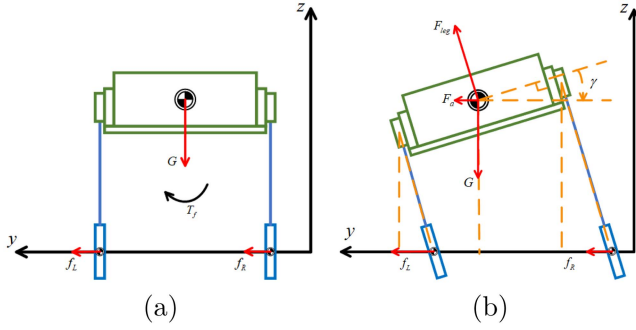


Fig. 4. The robot turns at a certain speed. (a) Axial friction produces a torque on the wheel. (b) Roll angle posture.

B. Control Strategy Optimization

1) *Steering Compensator*: In the base coordinate system $\{B\}$, the robot does not have the degrees of freedom to move laterally on its own. Therefore, lateral movement of the robot in the world coordinate system $\{W\}$ must be realized by the combination of acceleration, deceleration, and steering. When the robot turns at a certain speed, a centripetal force pointing towards the turning radius is required. The centripetal force required for turning is provided by the friction $f = f_L + f_R$. The maximum friction determines the upper limit of the steering speed. In real physical environments, because the point of force application is not at the center of mass, torque T_f will be applied to the body, as shown in Fig. 4(a). When the torque generated by the lateral friction exceeds the critical value of the torque acting on the body by partial gravity during steering, the robot may overturn, which is especially noticeable when the leg length is long, further limiting the steering speed. By actively shifting the robot's center of mass, as illustrated in Fig. 4(b). In this strategy, T_f is balanced by the torque generated by the horizontal component of the support force, and the critical moment of rollover is increased to enhance the stability of the robot during steering. According to the kinematics, the desired roll angle during the steering can be expressed as follows:

$$\gamma^* = -\arctan\left(\frac{\dot{x}\dot{\varphi}}{g}\right) \quad (20)$$

By analyzing the behavior of the bipedal wheeled-legged robot in controlling roll angle and leg length, it becomes apparent that a contradiction exists between these two aspects. This contradiction results in less accurate active control of the roll angle. Therefore, it is necessary to adjust l^* based on the roll angle appropriately. According to geometric relationships, it can be obtained:

$$\begin{cases} \Delta l_L^* = \frac{D}{2}\tan(\gamma) \\ \Delta l_R^* = -\frac{D}{2}\tan(\gamma) \end{cases} \quad (21)$$

As a result, when the robot is steering, changing γ^* and l^* can enhance the responsiveness of the robot's yaw angle.

2) *Gravity Compensator*: To simplify the lagrangian dynamics equations and the state-space equations, the effect of gravity is omitted from the part of generalized forces in the dynamics

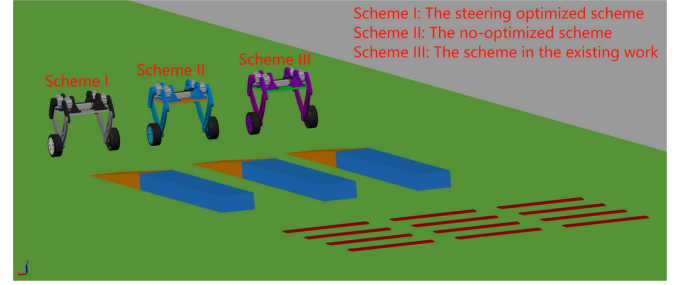


Fig. 5. Simulation environment. From left to right: scheme I, II and III.

equation. This omission will lead to insufficient normal force output from the left and right legs, causing the leg lengths to not reaching the set values. Moreover, during roll posture adjustments, the offset of the center of gravity can result in unequal forces on the left and right legs.

The compensator is required to perform feedforward compensation by calculating the normal force output required by the left and right legs to offset the gravity in rolling posture of the body. Through the force analysis, the torque balance equation can be obtained:

$$\begin{cases} F_{gL}\cos(\gamma) + F_{gR}\cos(\gamma) = m_b g \\ (F_{gL} - F_{gR})\frac{D}{2} + m_b g l_b \sin(\gamma) = 0 \end{cases} \quad (22)$$

The simplified result is:

$$\begin{cases} F_{gL} = m_b g \left(\frac{1}{2\cos(\gamma)} - \frac{l_b}{D}\sin(\gamma) \right) \\ F_{gR} = m_b g \left(\frac{1}{2\cos(\gamma)} + \frac{l_b}{D}\sin(\gamma) \right) \end{cases} \quad (23)$$

The gravity compensator can eliminate the influence of gravity on leg length in the modeling process, which greatly simplifies the complexity of modeling.

IV. EXPERIMENTAL RESULTS

A. Simulation

The Steering optimization scheme, the scheme before optimization and the scheme in the existing work are compared, which are referred to as scheme I, scheme II and scheme III respectively. Both schemes I and II adopt the same dynamics model and controller parameters. The matrix Q is set to $[1, 1000, 10000, 1, 8000, 1, 1000, 1, 3000, 1, 2000, 1]$ and matrix R is set to $[500, 500, 0.01, 0.01, 0.1, 0.1]$. The only difference between them is whether to use steering compensator. The parameters of balance-phase LQR in scheme III are also consistent with those in schemes I and II, and the PID parameters are set to match the performance of schemes I and II. The controllers in the three schemes have the same control frequency, and scheme III is used as the baseline for comparison with others. A physical simulation environment is constructed on a computer, as shown in Fig. 5.

1) *Dynamics Response*: The simulation results show that those three schemes have almost no difference in the dynamics response of linear acceleration on flat ground. As discussed in Section III-B, when steering at $t = 4$ s with $\varphi^* = 45$ degrees,

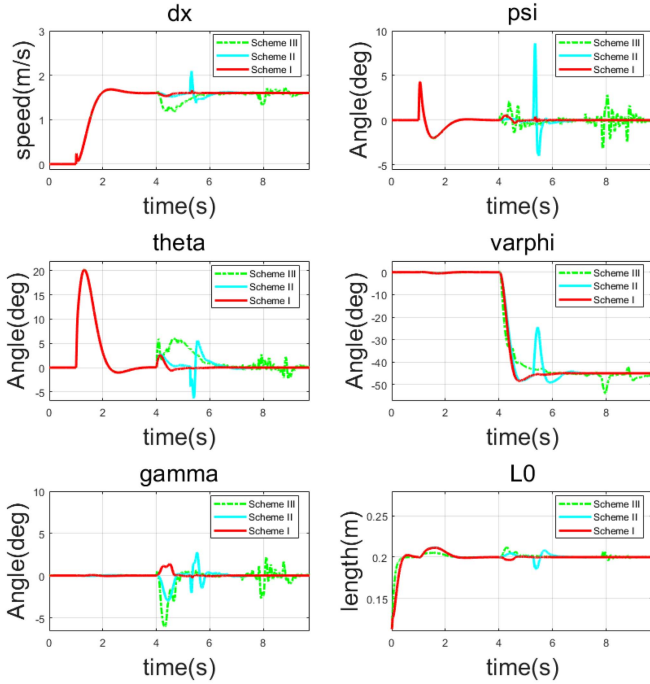


Fig. 6. The simulation results of the robots in the three schemes running on the flat ground.

scheme III exhibits noticeable incoordination jitter due to the decoupling control of some state variables. Scheme II, although capable of suppressing roll angle errors, has another issue when turning at 1.6 m/s. The excessive torque generated by friction leads to a tendency for the robot to overturn and has a substantial impact on all state variables, as shown in Fig. 6.

In contrast, Scheme I, by incorporating active roll angle control, successfully counteracts the tendency, effectively improving the dynamics response of the yaw angle and steering stability. In scheme I, the maximum speed during the steering can be increased to 3.5 m/s, representing a 118.7% improvement over the baseline.

2) *Terrain Adaptation*: To validate the effectiveness of the proposed model and optimized strategies, two representative terrains are selected for experiments : a single-sided bridge and a speed bump. The height of the unilateral bridge is 8 cm, and the height of the speed bump is 0.8 cm. The experimental results are illustrated in Fig. 7. From $t = 2.5$ s to $t = 5.3$ s, when the right leg crosses the single-sided bridge, Scheme I and II selectively suppresses the roll angle, and error is kept within 0.3 degrees. In contrast, Scheme III, which uses decoupled leg length variables and lacks appropriate compensation measures, results in a roll angle error approaching 2 degrees.

3) *Adaptive LQR for Interference*: In this experiment, an Adaptive LQR method is employed. The results are compared with the normal LQR algorithm. At $t = 1$ s, the interference is applied. The response are illustrated in Fig. 8. The experimental results indicate that the interference amplitude of each state variable is reduced by 17.0%, 23.2%, 24.5%, 17.0%, 9.4%, and 1.3%. Additionally, the setting times are shortened by 23.8%, 22.0%, -3.2%, 40.7%, 6.9%, and 5.9%. The dynamics response

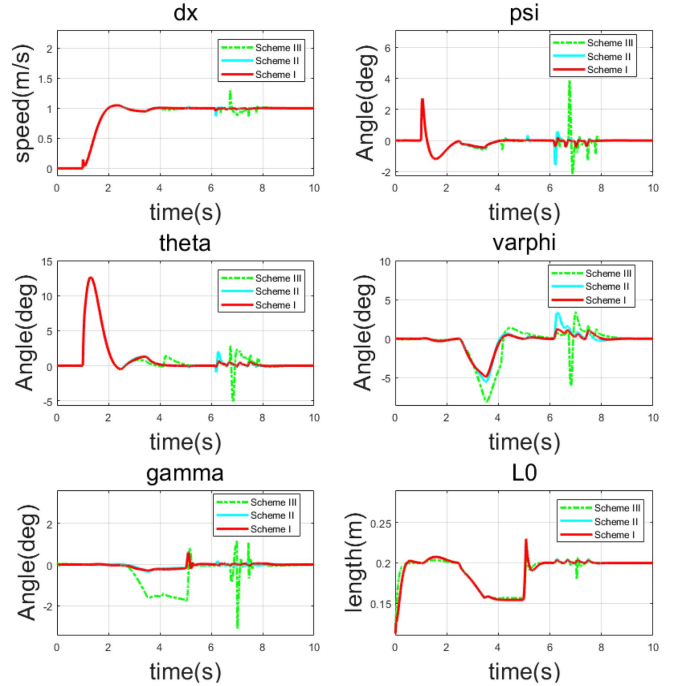


Fig. 7. The simulation results of the robots in the three schemes running on unstructured terrain.

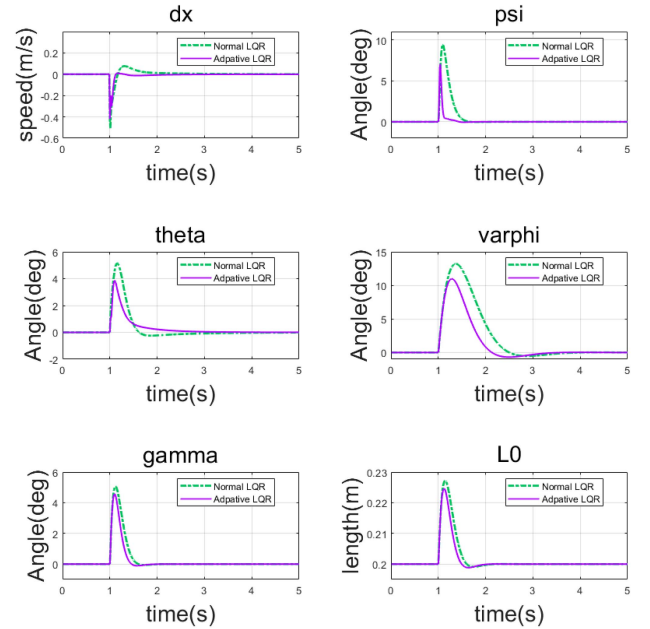
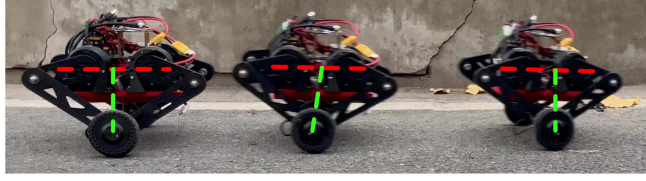


Fig. 8. The simulation results of impact resistance experiment of Adaptive LQR controller. The traditional LQR controller is used as the control group.

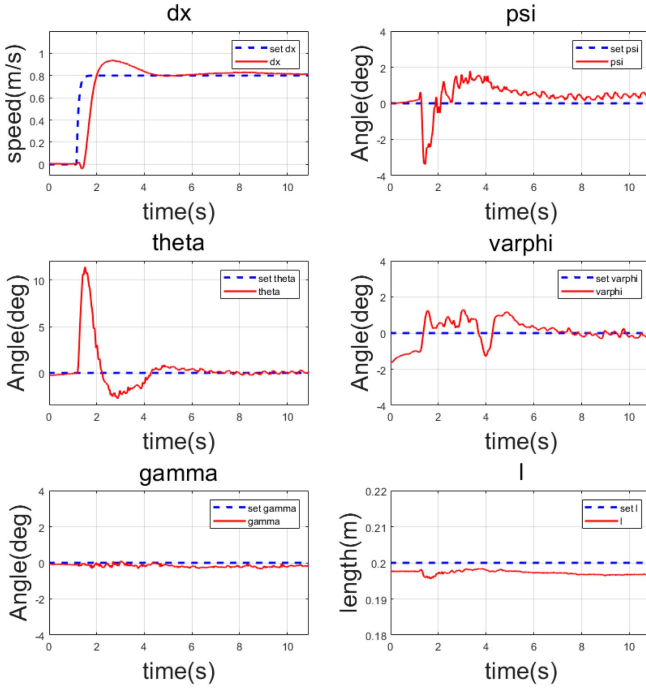
and stability of the robot show significant improvement. This improvement provides better stability for the installation of extended equipment.

B. Physical Experiments

1) *Robot Platform*: All parts of the robot are made of PLA material via 3D printing. The motors receive commands from



(a)



(b)

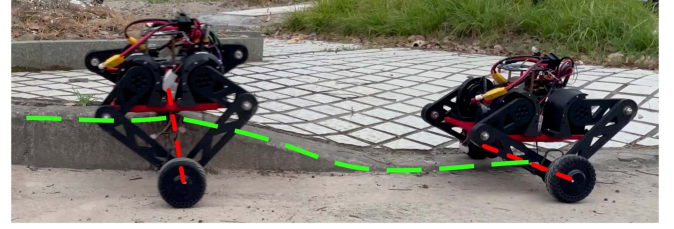
Fig. 9. The state response results of the prototype on the flat ground.

TABLE II
HARDWARE PARAMETERS AND ROBOT PARAMETERS

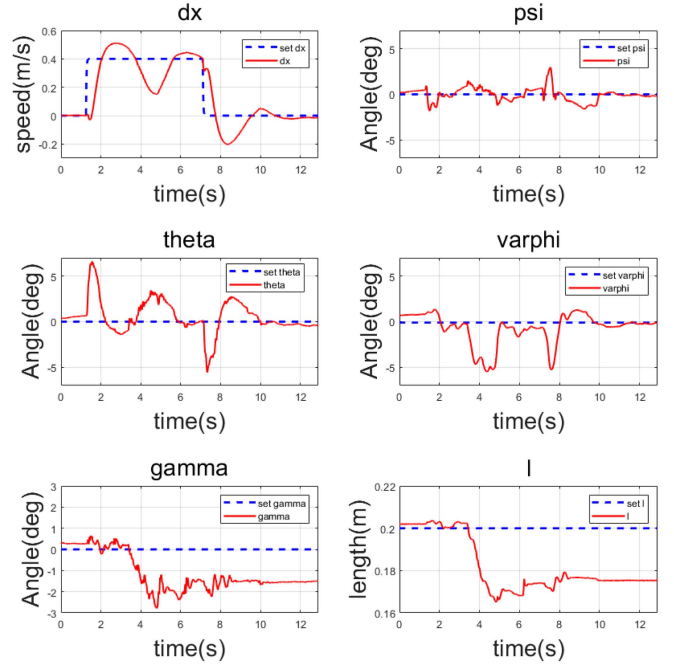
Component	Device model	Param	Value
Joint Motor	Xiaomi Cybergear	m_b	5.5kg
Wheel Motor	DDSM115	m_w	1kg
Onboard Computer	STM32H723VGT6	r_w	0.065m
IMU	MPU6050	l_1	0.1m
Battery	LiPo 10000mAh	l_2	0.2m

the MCU through the Controller Area Network(CAN) bus, while remote control is facilitated via Bluetooth, allowing for manual operation of the robot. The control board integrates an IMU with a maximum data output frequency of 200 Hz. Detailed hardware specifications of the robot and parameter value are provided in Table II. The Controller operates at a control frequency of 100 Hz.

2) *Dynamics Response*: The robot is tested in real environment. The process of the robot starting from a balanced position is illustrated in Fig. 9(a). Sensor data is collected via serial communication and plotted in Fig. 9(b). At $t = 1.1$ s, the speed reference $\dot{x}^*$ is set to 0.8 m/s. The overshoot σ is 16%, and the risetime t_r is 0.38 s. Compared to the simulation results, which



(a)



(b)

Fig. 10. The results of state response of the prototype on unstructured terrain.

showed an overshoot of 5% and a rise time of 0.65 s, there are discrepancies in the physical experiments. The differences can be attributed to the following factors:

In the simulation environment, there is no energy loss in the transmission mechanism of the robot, and the ground is completely flat. This idealization allows the body posture to be adjusted faster. In contrast, in the physical environment, The damping between the structures of the robot and the rough pebbles on the ground that slow down the adjustment of the body posture. These factors collectively impact the adjustment of various system states. The most noticeable effects are the increased overshoot and reduced risetime. To improve control performance, reducing damping in the transmission mechanism will be an effective way.

3) *Terrain Adaptation*: The curb with a sloped edge is a typical single-sided bridge terrain, as shown in Fig. 10(a). The robot runs on the bridge at a speed of 0.4 m/s and then stops on it. The results are shown in Fig. 10(b). It can be seen that the robot changes the length of its left and right legs after traveling on a single side bridge to maintain the balance. The roll angle error is controlled at around 2 degrees, the pitch angle error is controlled within 2.5 degrees, and the yaw angle error is controlled within 5

degrees. The experiment showed good control effect and verified the correctness of the theory.

V. CONCLUSION AND FUTURE WORK

In this article, a full state dynamics model, a trajectory function-based forward kinematics in VMC method, and an Adaptive LQR control method. The theoretical analysis and experimental results demonstrate that the proposed dynamics model, solution method, and control strategy can make the robot respond more quickly and smoothly through unstructured terrain. In the next work, we will optimize the prototype in many aspects. The biped gait model of the robot will be studied and established to overcome more complex terrain, such as stairs, ruins, etc. The robot will be equipped with ground information sensors such as LiDAR, and an obstacle prediction controller will be designed for Adaptive LQR control based on ground information.

REFERENCES

- [1] X. Liu et al., "Development of wheel-legged biped robots: A review," *J. Bionic Eng.*, vol. 21, pp. 607–634, 2024, doi: [10.1007/s42235-023-00468-1](https://doi.org/10.1007/s42235-023-00468-1).
- [2] Z. Mansor, A. Irawan, and M. Abas, "Evolution, design, and future trajectories on bipedal wheel-legged robot: A comprehensive review," *Int. J. Robot. Control Syst.*, vol. 3 no. 4, pp. 673–703, 2023, doi: [10.31763/ijrcs.v3i4.1107](https://doi.org/10.31763/ijrcs.v3i4.1107).
- [3] Official website of Boston Dynamics, "The evolution of stretch | Boston dynamics," Aug. 5, 2024. [Online]. Available: <https://www.youtube.com/watch?v=8WZoVJIV9V0>
- [4] X. Li, H. Zhou, S. Zhang, H. Feng, and Y. Fu, "WLR-II, a hose-less hydraulic wheel-legged robot," in *Proc. 2019 IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Macau, China, 2019, pp. 4339–4346, doi: [10.1109/IROS40897.2019.8967935](https://doi.org/10.1109/IROS40897.2019.8967935).
- [5] X. Li and Y. Fu, "Design and experimental evaluation of WLR-III," in *Proc. 2023 IEEE Int. Conf. Robot. Biomimetics*, Koh Samui, Thailand, 2023, pp. 1–6, doi: [10.1109/ROBIO58561.2023.10354666](https://doi.org/10.1109/ROBIO58561.2023.10354666).
- [6] V. Klemm et al., "Ascento: A two-wheeled jumping robot," in *Proc. 2019 Int. Conf. Robot. Automat.*, Montreal, QC, Canada, 2019, pp. 7515–7521, doi: [10.1109/ICRA.2019.8793792](https://doi.org/10.1109/ICRA.2019.8793792).
- [7] V. Klemm et al., "LQR-Assisted whole-body control of a wheeled bipedal robot with kinematic loops," *IEEE Robot. Automat. Lett.*, vol. 5, no. 2, pp. 3745–3752, Apr. 2020, doi: [10.1109/LRA.2020.2979625](https://doi.org/10.1109/LRA.2020.2979625).
- [8] S. Wang et al., "Balance control of a novel wheel-legged robot: Design and experiments," in *Proc. 2021 IEEE Int. Conf. Robot. Automat.*, Xi'an, China, 2021, pp. 6782–6788, doi: [10.1109/ICRA48506.2021.9561579](https://doi.org/10.1109/ICRA48506.2021.9561579).
- [9] L. Cui et al., "Learning-based balance control of wheel-legged robots," *IEEE Robot. Automat. Lett.*, vol. 6, no. 4, pp. 7667–7674, Oct. 2021, doi: [10.1109/LRA.2021.3100269](https://doi.org/10.1109/LRA.2021.3100269).
- [10] J. Yu, Z. Zhu, J. Lu, S. Yin, and Y. Zhang, "Modeling and MPC-Based pose tracking for wheeled bipedal robot," *IEEE Robot. Automat. Lett.*, vol. 8, no. 12, pp. 7881–7888, Dec. 2023, doi: [10.1109/LRA.2023.3322084](https://doi.org/10.1109/LRA.2023.3322084).
- [11] W. Du, M. Fnadi, and F. Benamar, "Whole-body motion tracking for a quadruped-on-wheel robot via a compact-form controller with improved prioritized optimization," *IEEE Robot. Automat. Lett.*, vol. 5, no. 2, pp. 516–523, Apr. 2020, doi: [10.1109/LRA.2019.2963822](https://doi.org/10.1109/LRA.2019.2963822).
- [12] Y. Xin, H. Chai, Y. Li, X. Rong, B. Li, and Y. Li, "Speed and acceleration control for a two wheel-leg robot based on distributed dynamics model and whole-body control," *IEEE Access*, vol. 7, pp. 180630–180639, 2019, doi: [10.1109/ACCESS.2019.2959333](https://doi.org/10.1109/ACCESS.2019.2959333).
- [13] X. Huang, T. Liu, M. Han, and J. P. Ferreira, "Virtual model control for dynamics balance of a two wheeled-legged robot," in *Proc. 2022 Int. Conf. Service Robot.*, Chengdu, China, 2022, pp. 31–37, doi: [10.1109/ICoSR57188.2022.00016](https://doi.org/10.1109/ICoSR57188.2022.00016).
- [14] K. C. Dinh, N. S. Dao, H. D. Le, H. L. Le, T. D. T. Cam, and D. T. Tran, "Kinematics and dynamics for a 4-DOF parallel robot," in *Proc. 2021 Int. Conf. System Sci. Eng.*, Ho Chi Minh City, Vietnam, 2021, pp. 151–157, doi: [10.1109/ICSSE52999.2021.9538416](https://doi.org/10.1109/ICSSE52999.2021.9538416).
- [15] J. Hu, "Vibration suppression of a high-speed flexible parallel manipulator based on its inverse dynamics," in *Proc. 2nd Int. Conf. Intell. Syst. Des. Eng. Appl.*, Sanya, China, 2012, pp. 744–747, doi: [10.1109/ISdea.2012.605](https://doi.org/10.1109/ISdea.2012.605).
- [16] J. Dong, R. Liu, B. Lu, X. Guo, and H. Liu, "LQR-based balance control of two-wheeled legged robot," in *Proc. 41st Chin. Control Conf.*, Hefei, China, 2022, pp. 450–455, doi: [10.23919/CCC55666.2022.9902200](https://doi.org/10.23919/CCC55666.2022.9902200).
- [17] Y. Zhuang et al., "Height control and optimal torque planning for jumping with wheeled-bipedal robots," in *Proc. 6th IEEE Int. Conf. Adv. Robot. Mechatron.*, Chongqing, China, 2021, pp. 477–482, doi: [10.1109/ICARM52023.2021.9536196](https://doi.org/10.1109/ICARM52023.2021.9536196).